

shop around. The opacity of muni markets means investors have to be especially vigilant—you will likely be ripped off and, given the poor transparency in the muni market, you won't even know it.

4. If you're a small investor, hold munis through low-cost mutual funds or ETFs (see chapter 16).
5. The MOB trade is not an "arbitrage." The same factors driving term spreads, such as volatility, are at play as in regular fixed-income markets. But there is a large added dose of illiquidity risk. See point 3 above.
6. Credit risk is likely to remain small. Of course, be smart about what kind of munis you buy. See point 3 again.

I close with a muni investment strategy. Many investors buying munis, including pass-through institutional investors like mutual funds, do so to avoid paying taxes. Munis are tax-exempt only at issue. Investors trading munis in secondary markets have to pay taxes. For example, if an investor purchases a par bond for a price less than par on the secondary market, he has to pay income tax on the difference between purchase price and the par value at maturity. This is taxation on *market discount* (as opposed to the no taxation on OID, which at issue is equivalent to tax-exempt coupon interest). Ang, Bhansali, and Xing (2010b) show that there are some very cheap munis with market discounts. Buying cheap market discount munis is like buying second-hand time shares. Everyone wants the new condo, but you get the exact same one for 50% less on the secondary market. Investors are reluctant to buy market discount munis because they are averse to tax—and many major muni mutual funds don't wish to pass on taxes to retail investors and thus avoid purchasing these bonds. The pricing anomaly is likely to persist because the muni market is fragmented: even if you were to know about this effect, you may not be offered the deals at cheap enough prices.

4. Tax-Efficient Allocation

The optimal asset allocation decision (ideally an optimal factor allocation decision, see chapter 14) concerns how much of each asset to hold. The optimal location decision is about where to hold these assets. The simplest choice is between a taxable or tax-deferred account. There are other asset locations: joint accounts with spouses and/or children, charitable vehicles, and many different kinds of trusts.

The optimal location problem is complicated because of the existence of *dynamic tax strategies*, and in some cases there are pure *tax arbitrages*. Some of these tax arbitrages are legal but permitted only to a certain extent, others are prohibited outright, and some are allowed but prohibited in "pure" forms and so fall into a murky area. For example, universities are prevented from issuing tax-exempt debt—they borrow at a rate lower than the federal government (see Section 3)—and then using that money to buy stocks. But, money is fungible. So the money in